

PART A

Experiment No.01

A.1 Aim: Identify a problem statement relevant to AI and describe its PEAS Descriptors with its Properties.

A.2 Prerequisite: Software Engineering, System Analysis, Understand different types of agent.

A.3 Outcome:

After successful completion of this experiment students will be able to

1. Understand, identify and analyse the problem, implement and validate the solution for the given task environment.
2. Applying fundamental engineering concepts appropriate to the discipline.
3. Potential to formulate and solve engineering problems in identifying the task environment.

A.4 Theory:

An agent is something that perceives and acts in an environment. The agent function for an agent specifies the action taken by the agent in response to any percept sequence. An agent is anything that can be viewed as perceiving its environment through sensors and acting upon that environment through actuators.

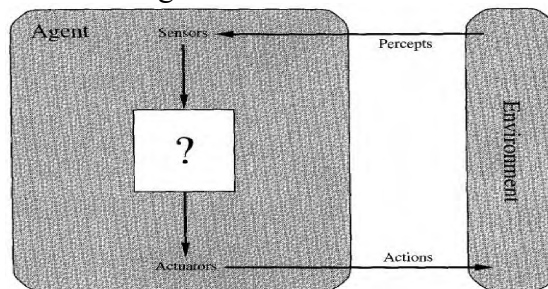


Figure: Agents interact with environments through sensors and actuators.

A performance measure embodies the criterion for success of an agent's behavior. When an agent is plunked down in an environment, it generates a sequence of actions according to the percepts it receives. This sequence of actions causes the environment to go through a sequence of states. If the sequence is desirable, then the agent has performed well.

A rational agent acts so as to maximize the expected value of the performance measure, given the percept sequence it has seen so far. A task environment specification includes the performance measure, the external environment, the actuators, and the sensors. In designing an agent, the first step must always be to specify the task environment as fully

as possible. Task environments vary along several significant dimensions. They can be fully or partially observable, deterministic or stochastic, episodic or sequential, static or dynamic, discrete or continuous, and single-agent or multiagent.

The task environment for an agent is comprised of PEAS (Performance measure, Environment, Actuators, Sensors)

PEAS specify the setting of an intelligent agent:

P: The performance measure defines degree of success.

E: What does the agent know about the environment?

A: The actions that the agent can perform.

S: Everything that an agent as perceived so far through its sensors.

Environment properties:

- **Fully (vs. partially) observable:** An agent's sensors give it access to the complete state of the environment at each point in time.

- **Deterministic (vs. stochastic):** The next state of the environment is completely determined by the current state and the action executed by the agent.

- **Episodic (vs. sequential):** The agent's experience is divided into atomic "episodes" (in which the agent perceives then performs one action), and the choice of action in each episode depends only on the episode itself.

- **Static (vs. dynamic):** The environment is unchanged while an agent is deliberating. (The environment is semidynamic if the environment itself does not change with the passage of time but the agent's performance score does.)

- **Discrete (vs. continuous):** A limited number of distinct, clearly defined percepts and actions.

- **Single agent (vs. multiagent):** An agent operating by itself in an environment.

Procedure/ Program:

Example of Agent types and their PEAS (performance, environment, actuators, sensor) Descriptions.

1] **English Tutor Example: PEAS**

Agent: Interactive English tutor

Performance measure: Maximize student's score on test

Environment: Set of students

Actuators: Screen display (exercises, suggestions, corrections)

Sensors: Keyboard

English tutor: Observable: Partially Agents: Multi (why?) Deterministic: Stochastic

Episodic: Sequential Static: Dynamic Discrete: Discrete

2] **Robot : PEAS**

Agent: Part-picking robot

Performance measure: Percentage of parts in correct bins

Environment: Convey p, or belt with parts, bins

Actuators: Jointed arm and hand

Sensors: Camera, joint angle sensors

PART B
(PART B : TO BE COMPLETED BY STUDENTS)

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Grade:	

B.1 Software Code written by student:

Problem Statement: Traffic Management System (AI Agent)

An AI-powered system designed to manage traffic flow in a smart city by controlling traffic lights, reducing congestion, and providing real-time traffic updates to drivers.

PEAS Descriptors:

Performance Measure:

- Reduction in traffic congestion and wait times at intersections.
- Improved average travel time for vehicles.
- Decrease in fuel consumption and emissions.
- Accurate real-time alerts for drivers about congestion or alternate routes.

Environment:

- Urban road networks with vehicles of varying types (cars, buses, bikes, etc.).
- Traffic signals, pedestrian crossings, and intersections.
- External factors like weather conditions, accidents, and road closures.
- Real-time data from traffic cameras, sensors, and GPS-enabled vehicles.

Actuators:

- Traffic lights and signal controllers.
- Digital traffic signs and notification boards.

- Alerts sent to navigation apps and devices.
- Emergency vehicle prioritization systems.

Sensors:

- Cameras and motion detectors at intersections.
- GPS data from vehicles.
- Inductive loop sensors embedded in roads to detect vehicle density.
- Data from weather stations and accident reports.

B.2 Observations and learning:

- Improved Decision-Making:
Accurate observations lead to better predictions and decisions.
- Adaptability:
Learning from observations helps the agent adapt to new or changing environments, such as unexpected traffic surges or evolving user preferences.
- Efficiency:
Effective learning ensures the agent uses resources optimally, reducing computational overhead and improving response times.

B.3 Conclusion:

The **PEAS** framework provides a systematic way to describe an intelligent agent, focusing on the essential components: its goals (performance measure), the environment it operates in, how it interacts with the environment (actuators), and how it perceives the environment (sensors). This helps in both designing and evaluating intelligent systems, ensuring that they are effective and aligned with their intended goals.

B.4 Question of Curiosity

(To be answered by student based on the practical performed and learning/observations)

Q1) Explain PEAS Descriptors in detail

PEAS is a framework used to describe the components of an intelligent agent:

1. Performance Measure (P): Defines how the agent's success is evaluated. It is the goal or criteria the agent tries to optimize.
 - Example: For a self-driving car, it could be safety, efficiency, and comfort.
2. Environment (E): The external context in which the agent operates. It includes everything the agent interacts with.
 - Example: Roads, traffic, and pedestrians for a self-driving car.
3. Actuators (A): The mechanisms that allow the agent to perform actions in the environment.
 - Example: Steering, brakes, and accelerator in a self-driving car.
4. Sensors (S): The components that allow the agent to perceive its environment and gather data.
 - Example: Cameras, radar, and GPS in a self-driving car.

Q2) Explain Current trends in AI.

- Generative AI: AI models like GPT (including ChatGPT) and DALL·E are revolutionizing content creation, enabling machines to generate text, images, music, and even code with human-like creativity.
- AI in Healthcare: AI is enhancing diagnostics, drug discovery, and personalized treatment plans. Applications include predictive analytics, medical imaging, and virtual health assistants.
- Autonomous Systems: AI is driving the development of self-driving cars, drones, and robots, making them more capable of operating in dynamic environments.
- Natural Language Processing (NLP): Advances in NLP, including transformers and large language models, are improving chatbots, translation services, sentiment analysis, and voice assistants.
- AI Ethics and Fairness: There is growing focus on addressing biases in AI models and ensuring transparency, fairness, and ethical considerations in AI development and deployment.

